Contributions at a glance

Bursting versus budding Ch. 6, 8

At matched mean output, bursting lowers extinction risk in the supercritical regime under linear clearance exactly when $L > 1$; offspring laws coincide at $L = 1$. With a positive integer removal budget on one delivery, bursting leaves more surviving output exactly when $L < 1$.

An exact two-type solution Ch. 7

Closed forms for the no-catastrophe probability and joint generating function, with one-way conversion between types and type-specific catastrophe rates.

The statistics of one bursting cell Chs. 4, 5

A unified calculation of load and release moments, burst timing and the burst-size/quasi-stationary identity, extended to several founders and successive cell-to-cell transfers.

From cells to infection dynamics Ch. 6

An age-structured model derived from single-cell laws. Fitted rates depend on the growth phase; one exponential phase cannot identify all three intracellular rates.

Kolmogorov's constant and the logistic map Ch. 3

Identifies $A(p)$ through the Koenigs function and proves that function hypertranscendental in its spatial argument. The closed-form question for $A(p)$ remains open.

Evolution under changing rates Ch. 9

A limiting species-count law and variance under capacity-dependent speciation, and an explicit moving critical point for competitive reversal in the Pimentel model.

Original statements, extensions and their published foundations are distinguished in *Originality of the results*, p. 269.